Q-BRIDGE: Verifiable Policy Queries over Joint Confidence Views

Zida Yang[1,†], Weiqing Zeng[2,†], Guquan You[3,†], Cong Wang[1,*], Jiexin Zheng[1,*]
[1]Guanghua School of Management, Peking University
[2]Southwest Jiaotong University
[3]Institute of Emergency Safety Innovations, Guangzhou University
2401111090@stu.pku.edu.cn, zengweiqing@my.swjtu.edu.cn, youguquan@gzhu.edu.cn
{wangcong,jiexinzheng}@gsm.pku.edu.cn
† Equal contribution. * Corresponding authors.

***Abstract*—Recommendation systems often use historical logs and randomized experiments to select policies, rules for assigning items to users, that improve outcomes over a reference. However, observational bias, sampling uncertainty, and limited transfer across populations make policy gains difficult to establish. Evaluating a limited set of candidate policies leaves a decision gap. When none guarantees the desired improvement, it remains unclear whether another policy could meet the target or whether the available evidence rules out such a guarantee for all policies. We treat these decisions as repeated queries over a reusable view of uncertain action outcomes, separating evidence construction from policy choice and query execution. Q-BRIDGE materializes observational intervals and randomized group constraints into a joint confidence view. It supports evaluating individual policies, finding the largest worst-case gain achievable by any policy, and certifying when no policy meets a requested threshold. For disjoint action groups, we develop a three-step algorithm that finds an optimal policy with independently verifiable bounds; exact linear programming handles more general constraints. Our algorithm matches the optimal guarantees computed by exact linear programming across 2,160 synthetic policy optimization problems. In 107 cases where evaluating the original candidate policies leaves the existence of a qualifying policy unresolved, searching the full policy family finds qualifying policies in 86 cases and rules out such guarantees in the remaining 21. Retaining exact certificate terms reduces repeated-query cost. Experiments on the Coat recommendation dataset illustrate how the available evidence can support policy redesign or justify stopping the search.**

***Index Terms*—query processing, uncertain data, materialized views, causal inference, verifiable computation**

I. Introduction

A recommendation analyst often reuses logs and randomized experiments to compare policies [1]. Policies assign actions to user groups; gain is expected improvement over a reference. The data stay fixed while the analyst changes the policy or required gain.

Production assignments may reflect unobserved preferences [2], and experimental results may transfer only at group level [3]. Sampling adds uncertainty. These sources permit several possible action-success probabilities. We seek a gain guaranteed across all possibilities consistent with the evidence and source assumptions.

Checking a candidate list leaves a decision gap. If no candidate meets the target, could another policy succeed, or does the available evidence rule out that guarantee for every policy? The answers call for policy redesign or stopping search on the unchanged view.

We develop Q-BRIDGE[1], a query framework that separates evidence construction from policy choice and execution. It materializes observational intervals and randomized group constraints into one *confidence view*. A policy is a linear query; its worst-case gain is the guaranteed answer. Q-BRIDGE also finds the largest guarantee over all policies and certifies when none meets a threshold. Each answer carries evidence checked independently against the stored view.

The data engineering problem. Repeated decisions raise two query-processing questions: which evidence remains valid when objectives change, and which work can be shared? Causal query systems address biased aggregates [4]; uncertainty-annotated databases preserve bounds across queries [5]. We connect these concerns through one jointly covered view, a whole-family query operator and exact certificate reuse. Adaptive queries reuse these bounds without rescanning events.

Running example. Figure 1 compares four jackets A–D for one population. Success means a high rating; $m_A, \ldots, m_D$ are the unknown rates. The reference selects each with probability $1/4$. Logs bound individual rates; transferred experimental group averages give $(m_A{+}m_C)/2 = .65$ and $(m_B{+}m_D)/2 = .70$. The reference's success probability is $(.65 + .70)/2 = .675$. Since $m_A \leq .6$, the first group forces $m_C \geq .7$. Always choosing C guarantees $.7 - .675 = .025 = 2.5$ percentage points (pp). The allowed vector $(.6, .7, .7, .7)$ limits every action to .7, ruling out a larger guarantee. Matching bounds proves optimality.

Figure 1 tests a margin $\delta = .001 = 0.1$ pp. Section IV-D works through these three policies to explain when saved evidence suffices and when a new solve is needed.

Materialize evidence once (II–III). Source summaries and declared assumptions define a shared view. Its joint coverage supports policies chosen after earlier query answers.

[1]Code and saved certificate examples: https://github.com/yzd2002/Q-BRIDGE-ICDE2027.

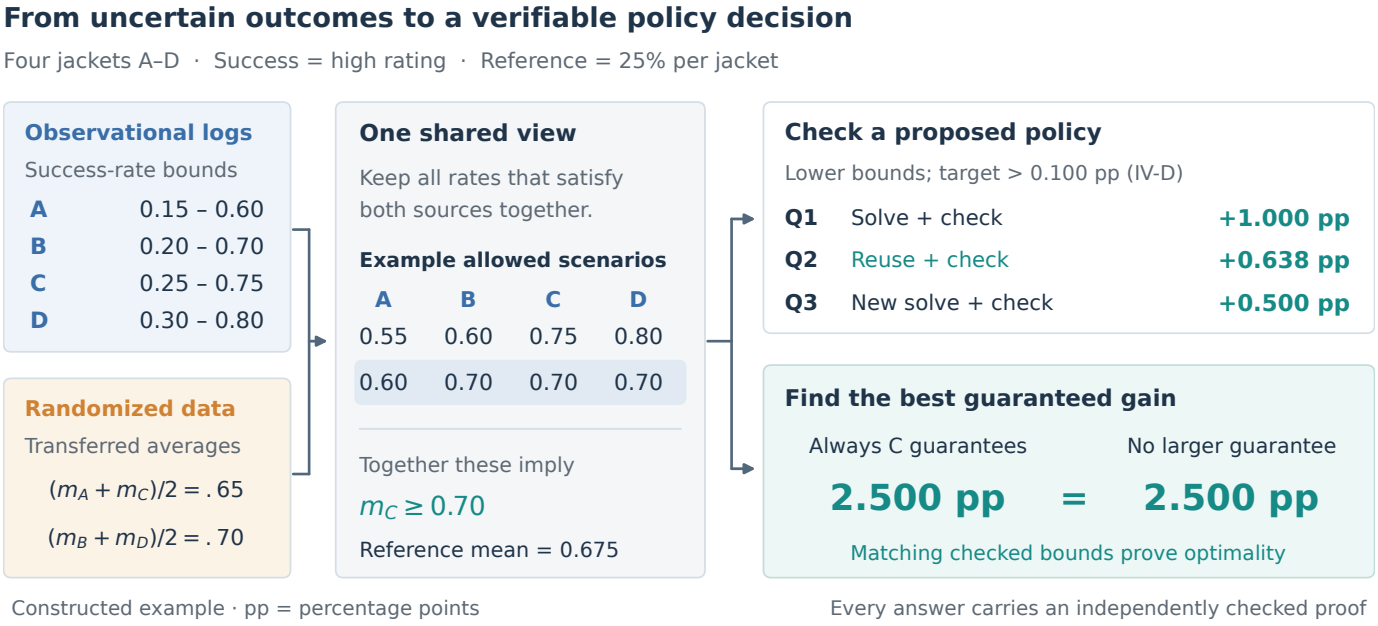


Fig. 1. One view, two query paths. Both sources constrain the same success rates; two allowed scenarios illustrate the view. The upper path checks three policies: Q2 reuses Q1's certificate; Q3 needs a new solve. The lower path establishes the best guarantee, 2.5 pp. This constructed example uses rounded inputs.

Answer beyond a candidate list (IV). For disjoint nonnegative groups, the smallest feasible cap yields an optimal policy and an original-view certificate. Exact LP handles overlaps.

Reuse checked work (IV–VI). Saved certificate terms answer changing objectives in linear arithmetic work. Controlled comparisons separate policy choice, source information and execution cost.

Table I summarizes the interface.

II. Policy Queries and Their Guarantees

We define guaranteed improvement, then the assumptions that turn each source into constraints.

A. The query and the stored object

Let $X \in \{1, \dots, K\}$ identify a user group, $A \in \{1, \dots, L\}$ an action such as recommending a jacket, and $Y \in \{0, 1\}$ success. There are $d = KL$ group–action pairs. The potential outcome $Y(a)$ asks whether this user would succeed if given action a, even when that action was not observed. Observational source O defines the target population; randomized source R may describe another population. Their means are $m^O_{xa} = \mathbb{E}_O[Y(a) \mid X = x]$ and $m^R_{xa} = \mathbb{E}_R[Y(a) \mid X = x]$.

Policy π_{xa} is the probability of taking action a in context x; π_0 is the reference and w_x the target context proportion. The expected gain is

$$\Delta(c) = \sum_{x,a} w_x(\pi_{xa} - \pi_{0,xa}) m^O_{xa} = c^\top m^O. \tag{1}$$

Thus $c_{xa} = w_x(\pi_{xa} - \pi_{0,xa})$ records where action probability increases or decreases. Each policy row and w sums to one; both are caller-specified.

Contexts group users; moment groups combine actions within a context. Index context–action pairs by $i = 1, \dots, d$. The view stores individual bounds ℓ, u and J group bounds a, b. Each row of T forms a weighted sum of action-success probabilities:

$$\widehat{\mathcal{C}} = \{m : \ell \leq m \leq u,\ a \leq Tm \leq b\}. \tag{2}$$

Each m is a possible success-rate vector. The bounds hold jointly: in Fig. 1, group averages couple the individual rates.

The *worst-case gain* is $V(c) = \min_{m \in \widehat{\mathcal{C}}} c^\top m$. A query tests $V(c) > \delta$, usually $\delta = .001 = 0.1$ pp. Under the source assumptions, Section III's construction contains all true target rates together with probability at least $1-\alpha$. Whenever it does, $V(c)$ lower-bounds actual expected gain.

Data and declarations. Events (x, a, y) are independent and identically distributed within each source; repeated observations of one user need user-level treatment (Section VI-D). Counts are N^s_{xay}, with $N^s_{xa} = \sum_y N^s_{xay}$ and $S^s_{xa} = N^s_{xa1}$. Randomization requires $A \perp Y(a) \mid X$ and known $q_{xa} = P_R(A = a \mid X = x) > 0$ on moment supports. A union bound allows dependence across sources and rows.

Prespecify contexts, target, source assumptions, rows, sensitivities and confidence allocation, optionally from independent design data. Policies and thresholds may then adapt on this view. A threshold is a required effect size: 0.1 pp means one additional expected success per 1,000 assignments, not a significance level. New snapshots or confidence constructions require a new view.

TABLE I
THREE POLICY QUESTIONS, WITH A DISTINCT CERTIFICATE FOR EACH ANSWER.

Question	What establishes the answer?	How it is obtained
Does this policy guarantee gain $> \delta$?	Guaranteed: a lower bound above δ. Not guaranteed: an allowed vector with gain at most δ. Undecided: neither established.	Reuse saved evidence or solve for this policy; overlapping groups are allowed.
What is the best guaranteed gain?	A policy lower bound matches an upper bound for all policies: their common value is capacity $\mathcal{R}$.	Three-step construction for disjoint nonnegative groups; exact LP for general groups.
Can every such guarantee be ruled out?	One allowed mean vector limits even the best policy's gain to δ.	Check a saved or newly found vector; full optimization is unnecessary if it suffices.

B. Bounding observational selection bias

The logged rate $p_{xa} = P_O(Y = 1 \mid x, a)$ describes users who received action a; m^O_{xa} describes what would happen if the target group received it. We bound their difference without estimating hidden preferences U. Let e_O denote assignment probabilities. Assume the recorded outcome equals the outcome under the assigned action, assignment is independent of potential outcomes given (X, U), and these probabilities are positive. Reweighting then gives

$$r^*(x, a, y) = \mathbb{E}_O\left[\frac{e_O(a \mid x)}{e_O(a \mid x, U)}\middle| x, a, y\right],$$
$$m^O_{xa} = \mathbb{E}_O[r^* Y \mid x, a]. \tag{3}$$

The weight r^* corrects over- or under-representation among logged successes and failures. The analyst declares $1/\Lambda \le r^* \le \Lambda$: $\Lambda = 1$ permits no such selection difference; larger Λ widens the bounds. This bounds an outcome-conditional average, not assignment odds or individual hidden-factor weights; logs alone cannot justify Λ [6].

For binary outcomes, these weight bounds translate directly into a permitted interval for the target mean [7, Eq. (9)]:

$$L_\Lambda(p) = \max\{p/\Lambda, 1 - \Lambda(1 - p)\},$$
$$U_\Lambda(p) = \min\{\Lambda p, 1 - (1 - p)/\Lambda\}. \tag{4}$$

Within one cell, write $p = p_{xa}$, $m = m^O_{xa}$ and $r_y = r^*(x, a, y)$. For $0 < p < 1$, the mean-one condition $pr_1 + (1 - p)r_0 = 1$ gives $r_1 = m/p$ and $r_0 = (1 - m)/(1 - p)$. Bounding both gives (4), exact for this relaxation [8].

Consistency provides additional information. Outcomes under a are known for those who received it; the remaining potential outcomes may be zero or one. Thus

$$P_O(A = a, Y = 1 \mid x) \le m^O_{xa} \le 1 - P_O(A = a, Y = 0 \mid x). \tag{5}$$

The denominator is everyone in context x. Both Fusion and observational-only (OBS) use these constraints. With exposure .8 and $p = .5$, they give $[.4, .6]$, versus $[.25, .75]$ from $\Lambda = 2$ alone.

C. Which randomized information transfers?

An experiment may support the target population's average rating probability for vests without establishing each vest's probability. Row j selects context x_j, actions S_j and nonnegative weights ν_{ja} summing to one. The normalized matrix H has $H_{j,(x_j,a)} = \nu_{ja}$ on this group and zero elsewhere. It is the exact row-normalization of stored T, fixed before inference (III-C). The declaration is

$$\tau_j = (Hm^R)_j, \qquad |(Hm^O)_j - \tau_j| \le \rho_j. \tag{6}$$

Here ρ_j permits source differences in a group mean; zero requires equality. Thus Λ controls selection bias, ρ transfer differences, and α sampling error. Two positive-weight action means can shift by $(h, -\nu_1 h/\nu_2)$ while preserving their average: group agreement does not establish action-level agreement.

If a policy contrast is a weighted combination of the transferred group averages, those averages determine its gain when known exactly: $c = H^\top v$ implies $c^\top m = v^\top \tau$ at $\rho = 0$. Other contrasts may remain uncertain. Assuming every individual item mean transfers ($H = I_d$, $\rho = 0$) is stronger; Coat includes this control.

III. BUILDING ONE CONFIDENCE VIEW

Group–action success/failure counts suffice for independent binary events: $O(n_O + n_R)$ aggregation work and $O(d)$ storage, with no later rescan. Repeated observations require user-level variances (Section VI-D).

A. Observational intervals and an explicit budget

Here a confidence *budget* is permitted failure probability, not sample size. Split $\alpha = \alpha_O + \alpha_R$ between sources. Of α_O, half covers d two-sided Clopper–Pearson (CP) intervals [9] $[p^-_{xa}, p^+_{xa}]$, each with failure probability $\alpha_O/(2d)$. The other half covers $2d$ one-sided joint-event bounds in (5), each at $\alpha_O/(4d)$, with denominator $N^O_x = \sum_{a,y} N^O_{xay}$. Denote their lower endpoints g^1_{xa}, g^0_{xa}. Then

$$\ell_{xa} = \max\{L_\Lambda(p^-_{xa}), g^1_{xa}\},$$
$$u_{xa} = \min\{U_\Lambda(p^+_{xa}), 1 - g^0_{xa}\}. \tag{7}$$

Monotonicity makes propagation conservative. Conditional binomial coverage permits random stratum counts. Empty action cells contribute $[0, 1]$; empty contexts add no joint restriction. The risk that any bound fails is at most the sum of their failure probabilities, even for dependent bounds.

B. Two prespecified randomized constructions

Prespecify one route and allocate α_R to it: average individual-mean intervals, or estimate group means directly. Both return $[\tau_j^-, \tau_j^+]$; transfer expands it by ρ_j, then III-C maps it to T_j. Synthetic primary comparisons use the first route; Coat reports both.

Projected cell intervals. Construct simultaneous binomial intervals $[r_i^-, r_i^+]$ for every randomized cell appearing in H, splitting α_R across those cells. Nonnegativity of H gives normalized group intervals $[Hr^-, Hr^+]$. Only *group* intervals enter (6); individual means need not transfer. Neither budget covers selecting the tighter route after observing data.

Group scores. Conditional on an event falling in (x_j, S_j), define $q_j(a) = q_{x_j a}/\sum_{b\in S_j} q_{x_j b}$ and $Z_j = \nu_{jA}Y/q_j(A)$. This reweights experimental assignment to the desired group weights: $\mathbb{E}[Z_j] = \tau_j$ and $0 \le Z_j \le B_j = \max_a \nu_{ja}/q_j(a)$. Let n_j count these events and $\bar{Z}_j$ be their average score. Counts, sums and squared sums follow from N^R_{xay}, since $Y^2 = Y$. If every multiplier $\nu_{ja}/q_j(a)$ equals one, $Z_j = Y$ and we use a binomial interval. Otherwise, for $n_j \ge 2$, the empirical Bernstein interval [10] has half-width

$$h_j = \sqrt{\frac{2\widehat{\mathrm{Var}}(Z_j)\log(4/\beta_j)}{n_j}} + \frac{7B_j\log(4/\beta_j)}{3(n_j-1)}. \tag{8}$$

Here $\beta_j = \alpha_R/J$, and the sample variance uses $n_j - 1$. The terms account for variation and range. The interval is $[\bar{Z}_j - h_j, \bar{Z}_j + h_j] \cap [0,1]$: only the expectation interval is clipped, not the scores. Insufficient observations use the full range. The union bound covers overlapping groups sharing events.

C. Keeping stored constraints faithful to the bounds

Stored bounds must not exclude rates allowed by the statistical intervals. The checker reads binary64 numbers as exact fractions: five stored values of .2 sum to $1 + 2^{-54}$, so even a group average requires explicit normalization.

For nonzero T_j, its exact mass $\kappa_j = \sum_i T_{ji}$ defines $H_{ji} = T_{ji}/\kappa_j$ and $T_j m = \kappa_j H_j m$; zero rows impose deterministic zero constraints. Assignment probabilities are either exact or explicitly normalized nonnegative design weights.

Scale by κ_j and round outward so storage cannot narrow the permitted interval:

$$\begin{aligned} a_j &= \mathrm{down}\big[\kappa_j(\tau_j^- - \rho_j)\big], \\ b_j &= \mathrm{up}\big[\kappa_j(\tau_j^+ + \rho_j)\big]. \end{aligned} \tag{9}$$

Projected intervals use exact $T_j r^-$ and $T_j r^+$ expanded by $\kappa_j\rho_j$. Observational endpoints, complements, score moments and normalization use rational arithmetic with one outward endpoint rounding; exact empty intervals are refused first. The statistical layer trusts CP quantiles and floating-point Bernstein arithmetic, including logarithms and square roots.

D. Why one view supports many queries

Set $\alpha_O + \alpha_R = \alpha$ (equal halves for fusion here). Once the true vector lies in the view, changing the policy changes only the objective; no new coverage event or confidence budget is needed.

Theorem 1 (One event, all objectives): Under the source and sampling conditions in Section II, $P(m^O \in \widehat{\mathcal{C}}) \ge 1 - \alpha$. On that event, every linear objective, including one chosen adaptively from previous answers, satisfies $V(c) \le c^\top m^O$. Any verified lower approximation to $V(c)$ shares this inequality. A method that cannot establish a usable view or bound makes no positive assertion.

Proof. Conditional on independent design information, interval coverage averaged over stratum counts and monotone mapping cover the box at $1 - \alpha_O$. Prespecified randomized intervals cover Hm^R at $1 - \alpha_R$; transport and outward mapping give target moments. A union bound covers m^O at $1 - \alpha$. On that event, minimizing over the view bounds every objective simultaneously, including adaptive choices. Weak duality extends this to checked lower bounds. □

Reuse retains the statistical design [11], [12]. A view identifier binds source-count digests, confidence allocation, assignment design, radius, tolerances and matrix. Changing these inputs invalidates reuse across views; changing only a policy or threshold preserves the shared evidence. The optimizer and checker therefore consume the same materialized object.

IV. Query Execution and Verification

The query engine returns a numerical answer together with a short, independently checkable proof, called a *certificate*. The optimizer produces it; a separate checker verifies it against the stored view $\mathcal{C} = \widehat{\mathcal{C}}$. This checks optimization; coverage follows from the source assumptions. We evaluate one policy (A), search all policies (B–C), then reuse proofs (D–E).

A. Checking a proposed policy

The checker first verifies a solver-proposed probability vector as a *feasible point*. Otherwise it marks the view `uncertain`, preventing decisions from contradictory constraints. Failure to find a point does not prove infeasibility, particularly for sets with no interior.

Together with a lower certificate, a checked feasible point brackets the optimum [13, Sec. 5.5.1]:

$$L(c) \le V(c) \le U(c) = c^\top \widetilde{m}. \tag{10}$$

A lower bound $L > \delta$ certifies the guarantee; $U \le \delta$ supplies a counterexample; otherwise the answer is unknown. These answers concern the guarantee, not the actual effect. Saved feasible points can bound later policies without another solve.

How a lower certificate works. Give the lower and upper group constraints nonnegative weights η_L, η_U. For any feasible m, both $\eta_L^\top(a - Tm)$ and $\eta_U^\top(Tm - b)$ are nonpositive. Adding them to the gain can only lower it. With $z = c + T^\top(\eta_U - \eta_L)$, minimizing this adjusted expression over the individual intervals gives

$$D(c;\eta) = a^\top\eta_L - b^\top\eta_U + \sum_i \min\{z_i\ell_i, z_i u_i\} \le V(c). \tag{11}$$

Each term chooses ℓ_i when $z_i \geq 0$, u_i otherwise. The weights form a *dual certificate*; a feasible vector is a *primal witness*. Both require only arithmetic to check. Any nonnegative weights are valid [14].

The checker uses exact arithmetic and rounds lower bounds downward, binding the objective, threshold and view digest. Failed or nonfinite proposals cannot authorize success. Independent optimization certificates are established [15].

B. Certifying a whole policy family

Testing one policy can miss a better one. The family Π contains all randomized policies on the available actions; its *capacity* is the largest worst-case gain. Fix a nonempty compact convex view $\mathcal{C} \subseteq [0,1]^{KL}$, exact probability weights $w \geq 0$, $\sum_x w_x = 1$, and reference $\pi_0 \in \Pi$. Stars denote returned optimizers. Define

$$\mathcal{R}(\mathcal{C}) = \max_{\pi \in \Pi} \min_{m \in \mathcal{C}} \sum_{x,a} w_x(\pi_{xa} - \pi_{0,xa}) m_{xa}, \tag{12}$$

$$F(m) = \sum_x w_x \left[\max_a m_{xa} - \sum_a \pi_{0,xa} m_{xa} \right]. \tag{13}$$

The inner minimum gives one policy's worst case; the outer maximum chooses the best policy. The reference ensures $\mathcal{R} \geq 0$. At any allowed m, choosing the best action per context gives $F(m)$, which upper-bounds each policy's worst-case gain. Thus $U(c)$ bounds one policy's guarantee; $F(m)$ bounds every policy's guarantee. A checked $F(m) \leq \delta$ excludes all above-threshold guarantees, not actual effects.

Compact convex domains and a continuous bilinear payoff permit minimax exchange [16], giving $\mathcal{R}(\mathcal{C}) = \min_{m \in \mathcal{C}} F(m)$. We therefore seek an allowed mean vector giving the best action as little advantage as possible, and a policy whose lower certificate reaches this upper bound.

C. Exact capacity for partitioned moments

The direct algorithm handles nonnegative, disjoint action groups within each context, including ungrouped actions. For each context, we seek compatible means that make the best action only slightly better than the reference. Write this gap as $t - p^\top m$, where t caps every action mean and $p = \pi_{0,x}$ is the reference distribution.

The algorithm in three passes. First find the smallest feasible common cap. Next raise the reference payoff as much as possible within that cap, obtaining an upper bound for every policy. Finally recover a policy whose checked lower bound equals it. In the running example, the cap is .7, the reference payoff is .675, and both bounds are .025. The cap is an intermediate device; the final certificate uses only the original constraints. Minimax and allocation are classical [17], [18]; our construction proves that the smallest feasible cap suffices and recovers an original-constraint certificate.

Theorem 2 (An attaining policy with an original-view certificate): Let $\mathcal{C}$ be a nonempty rational box-and-moment view with $0 \leq \ell \leq u \leq 1$. Suppose every nonzero moment row is nonnegative, belongs to one context, and has positive support disjoint from every other row. For any rational probability weights w and reference policy π_0, including zero entries, the three-step construction returns $m^* \in \mathcal{C}$, $\pi^* \in \Pi$ and nonnegative original-row multipliers η_L, η_U such that, with $c^*_{xa} = w_x(\pi^*_{xa} - \pi_{0,xa})$,

$$D(c^*; \eta) = c^{*\top} m^* = F(m^*) = \mathcal{R}(\mathcal{C}).$$

The dense implementation uses $O(dJ + d^2)$ exact arithmetic/comparison operations; the independent checker uses $O(dJ + d)$. These are unit-cost rational-operation counts, excluding coefficient-dependent bit costs, encoding and hashing.

Matching the policy lower bound and $F(m^*)$ verifies optimality. Unsupported rows are rejected; zero rows require $a_j \leq 0 \leq b_j$ and zero multipliers. Ties use index order.

Step 1: Find the smallest feasible cap.

The cap must leave enough success probability to meet each group's lower bound. Define

$$t_\ell = \max_i \ell_i, \qquad g_j(t) = \sum_i T_{ji} \min(u_i, t),$$
$$t_0 = \min\{t \geq t_\ell : g_j(t) \geq a_j \quad \forall j\},$$
$$h_i = \min(u_i, t_0).$$

Each g_j is piecewise linear between upper-endpoint breakpoints; its required root, together with t_ℓ, determines t_0. Upper group bounds cannot force a higher cap. Because the view is nonempty and groups are disjoint, each can choose mass in $[\max(a_j, T_j\ell), \min(b_j, T_j h)]$. Hence the capped view is feasible and its maximum is t_0. In Fig. 1, $t_0 = .7$ and $h = (.6, .7, .7, .7)$.

Minimum-cap sufficiency. A larger cap could raise the reference payoff, so feasibility alone is insufficient. For feasible m with $t = \max_i m_i$, clip to $v_i = \min(m_i, t_0)$, preserving lower boxes and upper moments. Restore violated lower moments within $[v, h]$ to $a_j \leq b_j$; disjoint supports prevent interference, giving feasible $v' \geq v$. Clipping lowers the maximum by $t - t_0$ and the reference average by at most that amount: $p^\top(m - v) \leq t - t_0$. Repairs only raise the average. Hence

$$\max_i v'_i - p^\top v' \leq t_0 - p^\top v \leq t - p^\top m.$$

Thus maximizing $p^\top m$ within this cap minimizes the original gap.

Step 2: Allocate means within the cap. With the maximum fixed at t_0, increasing the reference payoff reduces the gap. We allocate *means*, not policy probabilities. Raising m_i by ϵ costs group mass $T_{ji}\epsilon$ and gains $p_i\epsilon$. Starting at ℓ, fill each nonzero row in decreasing p_i/T_{ji} order on $T_{ji} > 0$, to total mass $\min(b_j, T_j h)$; uncovered coordinates take h_i. The added mass is $\min(b_j, T_j h) - T_j\ell$. Step 1 ensures the final total $\min(b_j, T_j h)$ meets a_j. Nonnegative weights justify filling even zero-ratio coordinates.

To build the certificate, record an upper-constraint multiplier λ_j per unit of group mass. Use zero when $b_j \geq T_j h$, including equality; the cap already limits the total. Otherwise use the largest support ratio if $b_j = T_j\ell$, or the last ratio receiving positive mass, including exact boundary exhaustion. After

subtracting these multiplier terms, $\xi = p - T^\top \lambda$ measures the remaining reference-payoff gain from raising each mean. Its sign explains the endpoint choice:

$$\xi_i > 0 \Rightarrow m_i^* = h_i, \quad \xi_i < 0 \Rightarrow m_i^* = \ell_i,$$
$$\lambda_j(b_j - T_j m^*) = 0.$$

A positive multiplier requires a tight upper bound. The example gives $m^* = (.6, .7, .7, .7)$, $\lambda = (0, 0)$ and family upper bound .025.

Step 3: Recover the policy and certificate. Choosing a best action at m^* alone is unsafe: B ties for best there, but allowed rates $(.6, .6, .7, .8)$ give pure B gain $-.075$. We must replace the artificial cap by a policy and original-constraint multipliers. Write the lower and upper multipliers as $\eta_L = \eta$ and $\eta_U = \lambda$. For equality in (11), the policy must select only maxima of m^*, rows with positive multipliers must be tight, and $z = \pi - p + T^\top(\lambda - \eta)$ must satisfy $z_i > 0 \Rightarrow m_i^* = \ell_i$ and $z_i < 0 \Rightarrow m_i^* = u_i$.

Start by canceling positive residuals at actions that can reach the cap; any missing probability will be supplied by a constraint that forces the cap:

$$\widetilde{\pi}_i = (\xi_i)_+ \mathbf{1}\{u_i \geq t_0\}, \qquad D_\pi = 1 - \sum_i \widetilde{\pi}_i \geq 0.$$

Here $(\xi_i)_+ = \max(\xi_i, 0)$. Since $0 \leq \widetilde{\pi}_i \leq p_i$, the deficit D_π is nonnegative. Initial coefficients $\widetilde{\pi}_i - \xi_i$ already have the required endpoint signs. In the example, initialization gives $(0, .25, .25, .25)$: the remaining .25 will go to C, the only action in the cap-forcing A/C group that can reach .7.

If $t_0 = t_\ell$, add D_π to a coordinate with $\ell_i = t_0$ and set all lower multipliers η to zero; its box coefficient increases at this lower endpoint. Otherwise choose the first row whose required root is t_0, so it forces the cap. Since $T_j h = a_j$, its supported means must equal h_i, and the zero-multiplier choice in Step 2 gives $\lambda_j = 0$, even if its upper moment is also tight. Set

$$E = \{i : T_{ji} > 0,\ u_i \geq t_0\}, \qquad \sigma_E = \sum_{i \in E} T_{ji} > 0,$$

where positivity follows because otherwise this row would attain its bound below t_0. Set $\eta_j = D_\pi / \sigma_E$, with other lower multipliers zero. Add $\eta_j T_{ji}$ to $\widetilde{\pi}_i$ on E. Updated box coefficients are zero on E and $-p_i - \eta_j T_{ji} \leq 0$ elsewhere in this row, where $m_i^* = h_i = u_i$; other coefficients are unchanged. The non-strict $u_i \geq t_0$ includes roots at breakpoints.

The completed policy sums to one and selects only maxima. The endpoint signs and tight rows with positive multipliers make $D(\pi - p; (\eta, \lambda)) = (\pi - p)^\top m^* = t_0 - p^\top m^*$. Multiplying each context's objective and both multiplier types by w_x and concatenating gives the theorem. Zero-weight contexts still need feasible means and legal policies. The checker validates original constraints, probabilities, weight signs and matching bounds; it never trusts the artificial cap. □

For the example's remaining mass, $\sigma_E = .5$, $\eta_L = (.5, 0)$ and $\eta_U = 0$, giving $\pi^* = (0, .25, .5, .25)$. The lower bound $.65(.5) - .6(.5) = .025$ equals the family upper bound $.7 - .675 = .025$. Pure C in Fig. 1 is another optimal policy. The example uses decimal inputs; measured runs check represented inputs.

Stopping without the full construction. Equal action means within each context make all policies tie with the reference, giving a cheap stopping certificate even with overlapping groups.

Proposition 1 (Zero capacity): If all $w_x > 0$ and $\pi_{0,xa} > 0$, then $\mathcal{R}(\mathcal{C}) = 0$ exactly when $\mathcal{C}$ contains a vector constant across actions within each context. Without strict positivity, a constant point is sufficient but need not be necessary.

Proof. The nonnegative summands of F vanish, under strict positivity, exactly at constant rows. Compactness attains its minimum; the minimax identity gives the result. □

Strict positivity is needed only for this shortcut's converse. With a sparse reference, $F(m) = 0$ allows unused actions to lie below the maximum: for $\pi_0 = (1, 0)$ and $m = (.7, .4)$, it is zero although the means differ. A failed constant-point test then falls through to the complete construction or exact LP, both of which allow zero reference probabilities; no smoothing is required.

For nonnegative single-context moment rows, even overlapping ones, form

$$I_x = [\max_a \ell_{xa}, \min_a u_{xa}] \cap \bigcap_{j : x_j = x} [a_j/\kappa_j, b_j/\kappa_j].$$

Here $\kappa_j = \sum_i T_{ji} > 0$; zero rows require $a_j \leq 0 \leq b_j$. Choose $v_x \in I_x$, set every $m_{xa} = v_x$, and check the original constraints. Under strict positivity, an empty intersection proves $\mathcal{R} > 0$, but not $\mathcal{R} > \delta$ for positive δ.

Saved policy lower bounds (including reference zero) and values $F(m)$ at saved feasible points also bracket capacity. They certify existence or exclusion if the corresponding bound crosses δ; otherwise search continues. Reuse requires identical $\mathcal{C}, w, \pi_0$. Section V-C separates loose bounds from restricted policy choice.

D. Reusing certificates across policies

A new policy changes only c, so saved nonnegative constraint weights still give valid, possibly loose lower bounds. The bank normalizes saved weights by their old objective's binary64 ℓ_1 norm; the new norm supplies scale s. We retain their group contribution $k = a^\top \eta_L - b^\top \eta_U$ and coefficient adjustment $r = T^\top(\eta_U - \eta_L)$. For stored $s \geq 0$,

$$D(c; s\eta) = sk + \sum_i \min\{(c_i + sr_i)\ell_i, (c_i + sr_i)u_i\}. \quad (14)$$

Thus later queries reuse the matrix–weight product r instead of recomputing it. Setup costs $O(\mathrm{nnz}(T) + d + J)$ exact operations, later checks $O(d)$; bit costs depend on coefficients. A fresh checker reconstructs the saved products exactly, using stored base weights and scale rather than separately rounded multipliers. Dual and bounded-Lagrangian reuse are established [19], [20], [21]; persistent LP reoptimization supplies a control [22].

The example: solve once, then try reuse. Q1 changes the uniform policy to $(.15, .15, .35, .35)$, giving $c = (-.1, -.1, .1, .1)$. The two group means imply

$$\Delta_1 = .27 - .2(m_A + m_B) \geq .01.$$

The allowed vector $(.6, .7, .7, .7)$ attains it; interval-only and group-only bounds are $-.075$ and $-.13$. Q1's weights $\eta_L = (.2, .2), \eta_U = 0$ give $k = .27$, $r = (-.1, -.1, -.1, -.1)$.

Q2 has $c = (-.105, -.09, .105, .09)$ and norm .39, versus Q1's .4. Scaling Q1's unnormalized weights by $.39/.4 = .975$ is equivalent to the bank's normalization. Equation (14) gives .006375, passing .001 without finding the optimum .0105. Q3 has $c = (-.05, -.15, .15, .05)$ and scale 1. Reuse gives $-.0225$, inconclusive; a new solve gives .005. A weak lower bound triggers work, not rejection.

E. Execution choices and numerical safeguards

Before solving a linear program (LP), the engine checks zero-weight bounds, last solver weights and feasible points. A first-in, first-out bank stores at most 16 candidates per enabled kind. Objective caching handles identical policies; the bank handles different ones. Zero-norm old objectives are not stored; a zero new norm uses scale zero. Solver cutoffs require exact checking, and undecided requests resume without a cutoff [23].

Two features target arithmetic cost. *Retention* saves exact terms k, r in (14). *Screening* uses cheap floating-point intervals to defer candidates that cannot decide the query. Screening never authorizes an answer; overflow triggers exact work. Before a false answer or a solver call, deferred candidates are replayed as needed to preserve the unscreened plan's evidence and solver path. Reuse hits retain the last solver state; zero-weight answers are not counted as reuse. The artifact specifies the replay order and cutoff handling.

Structural controls use box endpoints, continuous knapsack for single rows [24], or sparse HiGHS for overlaps [25]; only the reconstructed global dual is certified.

From stored contrasts to intended policies. For intended rational c^{ideal} stored as c, let $e = c^{\text{ideal}} - c$. Since $0 \leq m_i \leq 1$, the objective change lies between $\sum_i \min(e_i, 0)$ and $\sum_i \max(e_i, 0)$. Applying these corrections to the checked *minimum*-gain bracket gives

$$[L + \sum_i \min(e_i, 0),\ U + \sum_i \max(e_i, 0)]. \tag{15}$$

Apply the corrections before testing the ideal threshold, including at zero margin. Fixed-query timings (V-D) use stored contrasts, family queries (V-B) exact probabilities, and Coat (VI-D) both with (15) as needed.

V. Evaluation Design

The experiments isolate three data engineering questions: what changes when the policy family expands (VI-A), when the execution plan changes (VI-B), and when another source is integrated (VI-C)? The first two hold the view fixed; the third holds the candidate fixed. Synthetic worlds expose the true means, allowing answer correctness to be checked under controlled confounding and drift. Coat tests the complete source-to-decision workflow on real ratings (VI-D). Family analysis is retrospective; the source-information endpoint was registered before analysis. Exact-solver and Coat timings use an AMD EPYC 7763, Python 3.12.3 and one CPU/native thread.

A. Fresh worlds and policy design

A *world* fixes groups, assignment and success probabilities. The registered study draws 48 confounded worlds: $(K, L) \in \{(2, 4), (8, 8), (16, 16)\}$, uniform/skewed randomization and group drift $0/.02$, with four worlds per stratum and five inference samples per world. Twelve additional worlds test near-unit sensitivity. A latent binary factor affects exposure and rewards. Metadata $h_a \in [-.5, .5]$ have a designed positive reward association and are fixed before inference. Random groups permit individual shifts while controlling group drift.

Inference uses $2{,}000d$ O events and $100d$ or $1{,}000d$ R events. Separate design data ($500d$ O and $100d$ R events) choose three candidates before the inference data are seen. *Metadata* favors actions with higher fixed scores, using probabilities $\mathrm{softmax}(1.5h)$. *R heuristic* favors larger design-experiment success rates, using $\mathrm{softmax}(4\widehat{p}_x)$ with $\widehat{p}_{xa} = (S_{xa} + .5)/(N_{xa} + 1)$. Softmax normalizes exponentiated scores. *Robust design* selects the largest checked positive lower bound among twelve policy tilts, or keeps the uniform reference if none qualifies. Tilts use linear or alternating scores with slopes $\{-2, -1, -.5, .5, 1, 2\}$; ties retain the first maximum. Design outcomes never enter inference counts. We compare $\Lambda = 2$ with an oracle boundary computed from simulator truth and rounded upward. Counts and rules stay fixed; radius can change the design-selected policy.

Three sources supply constraints: *OBS* uses only logged data; *R* uses only transferred experimental group information; *Fusion* combines both. *Projected* averages action-level intervals; *group* estimates group averages directly (III-B). Each standalone method uses the full failure budget α; Fusion splits it between sources. The primary endpoint compares projected Fusion with certification by either full-α OBS or projected R, for the metadata policy, low R budget and $\Lambda = 2$. Average paired certification differences within each of 48 independent worlds. Their values lie in $[-1, 1]$, so the one-sided 95% Hoeffding lower bound is the mean minus $\sqrt{2\log(20)/48}$. The registered target requires this lower bound to exceed 0.05, i.e., a five-percentage-point increase in certification frequency. Other settings and the stratified t interval are descriptive. The standalone union counts a case when either source certifies it; it is not a jointly calibrated decision rule.

B. Whole-family queries on fixed views

We reuse the 48 main worlds with five samples each and target $\delta = .001$. Three source constructions (OBS, projected R, projected Fusion), two R sample sizes, two radii and finite/population information give 24 comparison cells. Population views remove sampling noise using the known simulator

statistics, while retaining uncertainty allowed by the source assumptions. The $48 \times 5 = 240$ sampled datasets yield 5,760 comparison positions. Both methods solve the 2,160 distinct exact $(\mathcal{C}, w, \pi_0)$ problems; all positions remain in comparisons.

The primary recovery study uses projected Fusion, $100d$ randomized events and $\Lambda = 2$. It searches the 107 of 240 finite views unresolved by saved evidence, keeping data and thresholds fixed. We first normalize and recheck the three saved policies exactly. This retrospective search uses Theorem 1's coverage for adaptive selection. We also read the same exact capacities at larger thresholds, without refitting or changing the 107-view subset.

C. A complete exact solver and the source of recovery

SoPlex 8.1.0 in rational mode [26], [27], [28] solves the epigraph LP on identical $(\mathcal{C}, w, \pi_0)$:

$$\min_{m \in \mathcal{C}, t} \sum_x w_x \big(t_x - \pi_{0,x}^\top m_x\big), \qquad m_{xa} \le t_x, \quad t_x \text{ free.} \tag{16}$$

Stationarity of free t_x gives $\sum_a \mu_{xa} = w_x$ for its nonnegative epigraph multipliers. Thus $\pi_{xa} = \mu_{xa}/w_x$ is a policy when $w_x > 0$; use the reference row when $w_x = 0$. Original-row duals and a feasible point complete the partition method's certificate contract. Import/export and checking are exact; neither the construction's answer nor a saved point is supplied.

A value-independent hash selects 78 requests for three paired rounds, alternating method order with fresh models and workers. Service time covers transmission after worker initialization through certificate checking, including serialization; startup/cleanup are reported separately. Limits are one CPU thread, ten seconds per service and 8 GiB address space. Failures retain their costs and denominators. Repeated rounds measure machine variation, not new datasets.

To determine whether recovery comes from better bounds or a different policy, we solve the original three policies exactly, obtaining optima L_i^*. Set $L_3^+ = \max(0, L_1^*, L_2^*, L_3^*)$, with zero from the reference. Here L_{saved} is the maximum of zero and saved lower bounds for these same three policies. Then

$$\mathcal{R} - L_{\text{saved}} = \underbrace{L_3^+ - L_{\text{saved}}}_{\text{certificate-transfer gap}} + \underbrace{\mathcal{R} - L_3^+}_{\text{three-policy restriction}} . \tag{17}$$

Both terms are nonnegative. Exact policy normalization can loosen saved bounds; missing optima leave the decomposition unknown. The recovery set and threshold stay fixed.

Coat comparisons. Item gender, jacket type and color define separate partitions. We also test their combined, overlapping constraints. Subviews of one parent retain its bounds and inherit its joint coverage event. Across distinct parent constructions or assumption declarations, no simultaneous 95% guarantee is asserted.

D. Repeated fixed-policy queries

The 63 inputs comprise three primary 1,000-query streams on a 512-cell view; 54 geometry streams varying size ($d = 64, 512, 2048$) and row overlap; and six generated-log streams with 64 actions, one million O and four million R events. The latter two groups contain 100 queries per stream. Queries vary nearby and unrelated policy directions at $\delta = .001$. Primary streams have one solve and 999 reuse hits, all true; others include false and unknown answers. These controlled workloads do not estimate online hit rates.

Four plans toggle screening S and retention P, using identical solvers, candidate order and checker; answers, evidence and solver paths must match. The optimized implementation retains decoded views, feasible points and within-query number conversions, and skips zero coefficients. Thus the retention comparison isolates saving the exact terms in (14). A separate candidate-order control checks the last solver certificate first; changing this order can change the returned certificate.

Service includes initialization, validation, all queries and fallback, serialization and checking with a 16-entry bank; it excludes ingestion, disk I/O, compression, network transfer and archival audits. A quality pass checks 45,000 responses; five shuffled timing rounds contribute 225,000 more. Each fresh run uses one CPU/native thread. Ratios compare enabled/disabled cost on the same input with the other switch fixed; group summaries give each input equal weight in a geometric mean. Rounds reflect machine variation. Individual costs, slow cases and all four conditional comparisons are archived.

VI. Experimental Results

A. What complete family optimization adds

Broader policy search changes the fixed-view answer. Of 240 primary views, earlier checks find a qualifying policy in 12 and exclude every qualifying policy in 121 (zero capacity). Full-family search resolves the remaining 107: it finds a different qualifying policy in 86 and rules out the target in 21, whose best guarantee is positive but at most δ. Thus 98 views support the target and 142 do not. Exact reoptimization of the original three policies recovers none of the 86: their best value, including the reference, is zero in every case. Recovered guarantees have median 0.991 pp, interquartile range $[0.583, 1.549]$ pp and range $[0.105, 3.632]$ pp. At thresholds 0.5 and 1.0 pp, 71 and 42 of the same 107 views admit a qualifying policy, respectively; the original candidates still support none. This post hoc threshold readout changes no view or outcome. Policy choice explains the matched-view difference; gain magnitudes describe the simulation.

The construction and exact LP agree on all 2,160 distinct synthetic problems. Across family, fixed-policy, Coat and timing checks, all 12,770 responses pass, without timeout or incomplete call.

Exact answers with dimension-dependent cost. Across three rounds, construction/SoPlex geometric-mean service ratios are 0.808–0.835 at $d = 8$, 1.066–1.129 at $d = 64$, and 1.170–1.196 at $d = 256$. The construction is slower in 5/78, 55/78 and 54/78 pairs, respectively. These compare checked Python rational construction with compiled rational LP. Table II gives absolute costs. At $d = 256$, verification alone takes median 144.3/144.6 ms (Part/LP); certificates occupy about 149.5 kB.

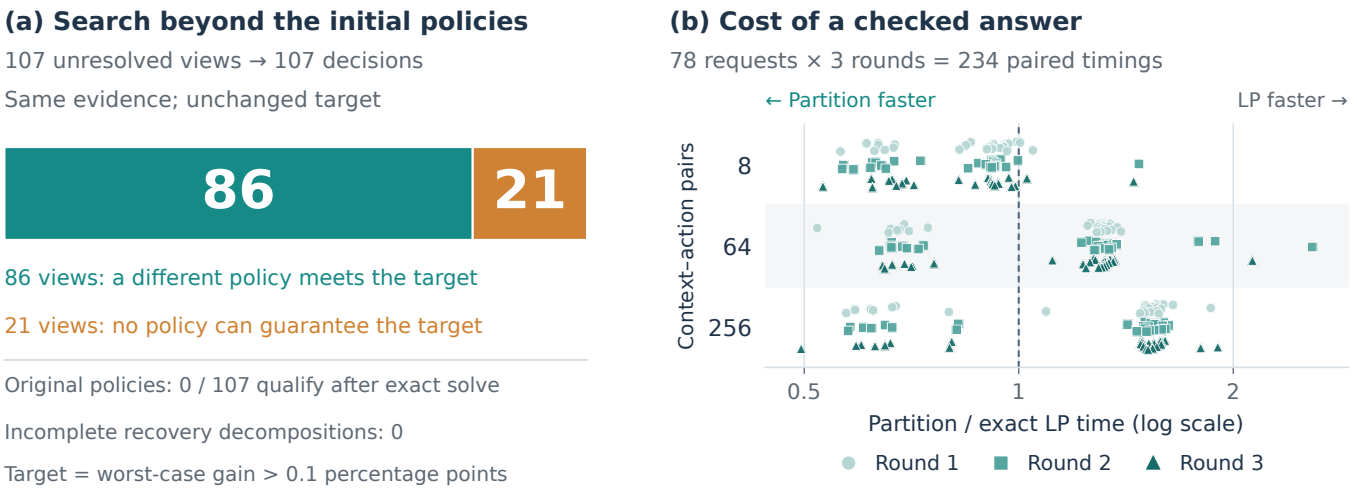


Fig. 2. Full-family search and its cost. (a) Search resolves the 107 remaining views from 240 primary cases. Recovery requires a different policy. (b) All answers verify: 26 requests per dimension, three dimensions, three timing rounds. Response time includes checking; vertical offsets separate rounds and requests. The dashed line denotes equal cost; rounds measure machine variation.

TABLE II
ABSOLUTE COST AND PROOF SIZE OF A CHECKED ANSWER.

d	Method	Answer (ms)	Check (%)	Total (ms)	Proof (kB)
8	Part	4.66	24.4	122.2	3.47
8	LP	4.99	22.1	135.2	3.47
64	Part	53.58	23.2	172.2	20.93
64	LP	40.85	31.6	170.0	20.91
256	Part	648.13	22.8	770.3	149.51
256	LP	419.97	34.6	554.0	149.44

Separate medians over 26 requests × 3 rounds per method/dimension. d: context–action pairs; Part: partition construction; LP: SoPlex. Answer: worker readiness through checking; Check: verification as a percentage of answer time; Total: includes startup/cleanup; kB: 1,000 bytes. Paired ratios: Fig. 2(b).

Source comparison on family capacity. At the larger R budget with conservative radius, projected R passes 223/240 thresholds versus Fusion's 218/240; R has greater capacity in 128 pairs, Fusion in 100, with 12 ties. At the smaller R budget with oracle-boundary radius, OBS and Fusion both pass 239 thresholds, but OBS has greater capacity in 219 pairs and Fusion in 21. Fusion protects both sources within the same total error budget, so its component intervals can be wider; VI-C separates this cost from the gain due to added constraints. Generating means satisfy all 2,880 finite partition views, and every returned policy assertion agrees with stored truth.

B. The cost of answering repeated policy queries

Retention saves the exact group terms that a new query would otherwise recompute; screening first tests whether an exact check can be deferred. All 270,000 responses pass independent exact replay; core plans match evidence, decisions and native paths. Across the five primary rounds (Fig. 3), retention ratios are 0.735–0.748 without screening and 0.766–0.791 with it. Screening ratios are 1.103–1.138 without retention and 1.165–1.186 with it. Screening also costs more in every geometry and source-derived group-round mean. Its earlier benefit without retention disappears under optimized arithmetic. Retention reduces cost with either screening setting in all five primary rounds.

The candidate-order control costs 0.833–0.882 times S_0P_1 on primary streams, but is slower in 1/15 primary, 66/270 geometry and 17/30 source-derived input-round pairs. Retention also has adverse individual cases despite lower group means. Thus retention lowers group-average cost without guaranteeing an individual speedup. Only this control changes candidate order.

a) Auxiliary execution controls: Additional tests check stronger efficiency claims. Screening plus retention does not reach the registered cost-ratio target ≤ 0.8 against each feature alone; screening slows every broader-group round. A larger bank also misses its speed target and changes decisions. Materialization speeds ingestion, but preparation costs 1.580/1.619 times hot execution under uniform/skewed assignment. Full controls are archived.

C. Source information and query guarantees

Table III follows three policies chosen before inference (48 worlds $\times 5$ samples each); VI-A instead searches for a better policy on each fixed view. All 1,200 inference configurations across 60 worlds complete. For the metadata policy at $100d$ randomized events and $\Lambda = 2$, fusion certifies 12/240 versus six for the standalone union (OBS six, R zero): six additions, no losses, certification-rate difference 0.025 (2.5 pp). The one-sided Hoeffding lower bound, -0.3283, does not meet the required 0.05 increase; the descriptive t interval is $[-0.0249, 0.0749]$. These are changes in certification frequency, not policy gains. All six additions occur in two zero-drift 2×4 worlds.

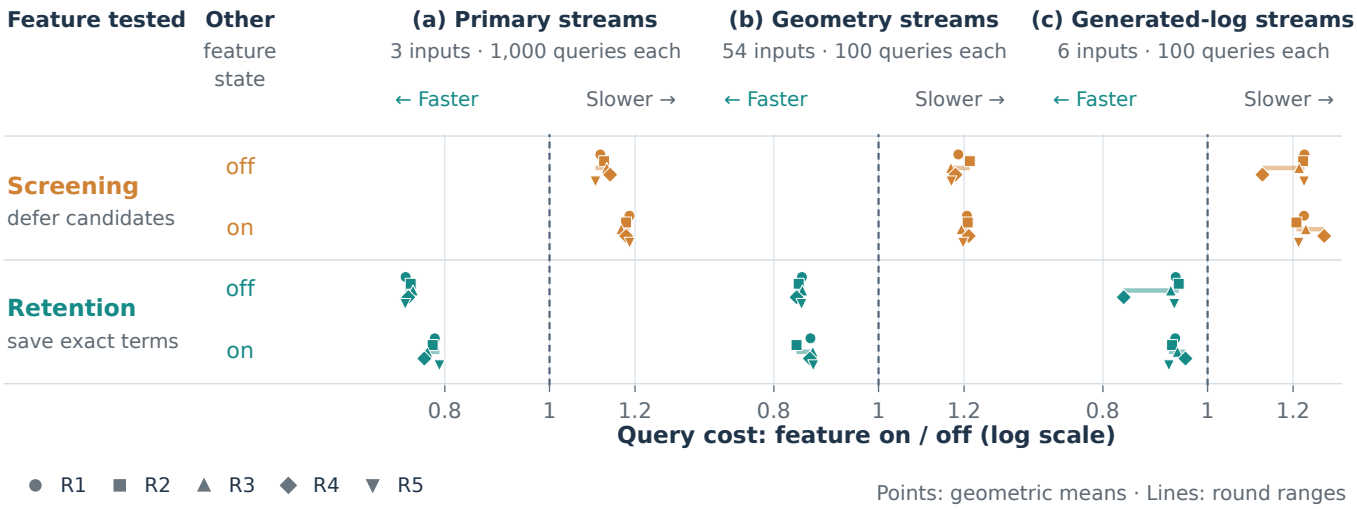


Fig. 3. Retention lowers group-average cost; screening does not. Rows toggle the named feature while fixing the other off or on. Each point is one timing round's equal-input geometric mean of on/off cost ratios (R1–R5). Segments span the five rounds, not confidence intervals. The dashed line marks equal cost; below one is faster.

TABLE III
PROJECTED FUSION: EACH ROW PARTITIONS 240 QUERIES.

Policy	Certified	Limited by sampling	Limited by ambiguity	True gain $\leq \delta$
Metadata	12	33	195	0
R heuristic	4	43	190	3
Robust design	2	0	0	238

240 queries per policy with $100d$ randomized events, $\Lambda = 2$, and $\delta = .001$ (0.1 pp). Sampling-limited: exact population statistics would suffice. Ambiguity-limited: even those statistics do not suffice despite true gain $> \delta$. "Robust design" selects one candidate before inference; it is not full-family optimization. No case is missed solely by optimization.

Larger-budget projected fusion certifies 15 metadata, 16 heuristic and two robust-design queries; transfer and radius uncertainty remain.

Why adding a source need not improve the bound. For the same policy contrast c, fusion adds group constraints but splits the confidence budget. Let $\mathcal{B}_\alpha \subseteq \mathcal{B}_{\alpha/2}$ be full-budget OBS and fusion's observational box, with objective minima $B_\gamma = \min_{m \in \mathcal{B}_\gamma} c^\top m$. For nonempty $\mathcal{C}_F = \mathcal{B}_{\alpha/2} \cap \{a \leq Tm \leq b\}$, its optimum obeys

$$V_F - B_\alpha = \underbrace{(V_F - B_{\alpha/2})}_{G \geq 0} - \underbrace{(B_\alpha - B_{\alpha/2})}_{P \geq 0}. \tag{18}$$

Fusion improves the full-budget observational bound exactly when its constraint gain G exceeds its budget cost P.

Proposition 2 (A checkable zero-gain condition): Let $F_c = \arg\min_{m \in \mathcal{B}_{\alpha/2}} c^\top m$. Then $G = 0$ if and only if F_c contains a point satisfying all randomized group constraints. If F_c is a singleton, checking its moment rows decides whether $G = 0$ or $G > 0$.

Proof. A feasible box minimizer attains the box bound in $\mathcal{C}_F$. Conversely, compactness gives an optimizer in $\mathcal{C}_F$; equality places it in F_c. If the sets do not intersect, the gap is strict.□

Choose box endpoints by the signs of c and check the group rows. Feasibility proves zero constraint gain. If zero coefficients leave multiple minimizers, one failed point is inconclusive. We apply this test below.

Sampling uncertainty versus ambiguity. Finite views include sampling intervals; population views use exact generating statistics, retaining ambiguity. For the metadata policy, all 240 true gains exceed δ: 12 queries are certified, 195 are limited by ambiguity and 33 by sampling uncertainty (Table III). Exact optima exclude optimization failure. Six additional finite certifications comprise four with sufficient standalone population information and two requiring fusion. Robust design retains the reference in 238/240 lower-budget cases: true effect zero.

For the metadata policy, Fusion adds/loses 6/0, 9/0, 0/2 and 3/1 for $(\Lambda = 2, 100d)$, $(2, 1000d)$, $(\Lambda^*, 100d)$ and $(\Lambda^*, 1000d)$, respectively; oracle Λ^* comes from simulator truth. In the low-budget oracle case, constraint gain averages 0.0114 pp versus 0.0913 pp budget cost. The zero-gain test holds for 193/240 queries, including both losses; the larger-budget loss has positive gain below cost. All three losses persist at exact optima. Different samples and design policies prevent a causal sample-size interpretation; full diagnostics are archived.

a) Standalone-source controls: Observational consistency (5) lets OBS certify 99/100 external 2×2 queries; other policy/view settings differ. KuaiRec's imposed exposures and query-directed randomized means [29] do not establish general complementarity: a 35-setting census finds no true-radius fusion-only case. 400 count/policy pairs remain in the artifact.

Additional checks cover three objects: 6,000 inference views have 14 coverage misses (outside the 2,880-view family subset) and no false certification; 18,900 final-mapping proofs

all verify, leaving seven group-score certifications unresolved; 600 design views have one coverage miss.

D. Real data: policy redesign and stopping

Coat supplies 24 self-selected and 16 random ratings per user, for 290 users and 300 items [1]. It tests the source-to-decision workflow for allocating one item recommendation in a population ($K = 1$). Success means a rating of at least four; zeros are unobserved entries. Users, not the 11,600 ratings, are independent; potential ratings are assumed unaffected by panel order or other presented items.

The decision is to stop searching or redesign. At primary $(\Lambda, \rho) = (2, 0)$, all five views have zero capacity: another candidate cannot establish a positive guarantee from those views. At the stronger declaration $(1.5, 0)$, group Fusion supports a new policy with checked gain 0.246355 pp. Unchanged ratings and a new declaration define a different view, making the evidential requirement explicit.

Group constraints use *item* attributes: gender category, jacket type and color. Three partitions total 31 rows, overlapping when combined. Twelve assumption settings cross $\Lambda \in \{1, 1.5, 2, 3\}$ with $\rho \in \{0, .02, .05\}$; the primary setting is $(2, 0)$. Fifteen candidate policies upweight or downweight front-page, men's, fleece, waterproof or black items: $\pi_i \propto t^{h_i}$ for binary attribute h_i and $t \in \{.5, 1.5, 2\}$. The reference selects uniformly $(1/300)$; the primary candidate doubles front-page item weights, then normalizes. Only the three front-page policies have contrasts outside the span of group rows.

Using all ratings assumes each user's 16 R items are a simple random subset independent of potential ratings. The release describes random/uniform selection; collection code is unavailable. A control keeps one value-blind rating per source/user, with $q_a = (16/300)/16 = 1/300$. Its 580 events leave 116 empty O and 124 empty R cells; none of 900 queries or 15 full-cell controls is certified.

Inference keeps users as replicates. O intervals count included users per item; joint bounds use $N = 290$. Uniform thinning divides inclusion probabilities by 24, preserving their ratio and Λ while weakening consistency bounds. For R groups, each user with a sampled item in that group contributes one within-group average. The resulting 72–290 independent user scores are unbiased under the subset assumption, lie in $[0, 1]$, and enter (8) with variance denominator $n_j - 1$. Projected intervals instead propagate per-item CP through (9).

Five constructions across twelve settings, plus one full-cell control, give 61 views and $15 \times 61 = 915$ candidate queries. No view is refused or query certified. Under $(\Lambda, \rho) = (2, 0)$, group Fusion improves R group's lower bound by 0.309 pp, but neither passes $\delta = .001$.

In Table IV, Fusion group's -1.193 pp bounds one policy's gain, whereas $4/12$ counts assumption settings supporting some qualifying policy, all outside the primary setting. Independent checks reconstruct the statistics, views and proofs.

Stopping under the primary declaration. Group Fusion allows every item to have success probability 0.2186 (rounded), making every policy tie with the reference. Further search cannot establish improvement on this view; this does not assert equal real preferences. The stronger full-cell control also has zero capacity.

TABLE IV
COAT: ONE POLICY VERSUS THE FULL POLICY FAMILY.

	Fixed policy Primary setting	Full policy family Assumption settings		
Construction	**LCB (pp)**	**Zero**	**Small**	**Qualifies**
OBS	-3.729	9/12	0/12	3/12
R group	-1.502	12/12	0/12	0/12
R projected	-3.402	12/12	0/12	0/12
Fusion group	-1.193	7/12	1/12	4/12
Fusion projected	-3.132	9/12	3/12	0/12
R full-cell control	-3.077	1/1	0/1	0/1

LCB: one fixed policy at primary $(\Lambda, \rho) = (2, 0)$, in pp. Family columns: $\mathcal{R} = 0$, $0 < \mathcal{R} \le \delta$, or $\mathcal{R} > \delta$ across twelve settings on the same 290 users. All five primary capacities are zero. Full-cell transport is a stronger, one-setting control. Here $\alpha = .05$, $\delta = .001$ (0.1 pp).

Searching beyond candidates under other declarations. With ideal uniform reference $1/300$, exact family optimization resolves 60 grid positions: 49 have $\mathcal{R} = 0$, four $0 < \mathcal{R} \le .001$, and seven $\mathcal{R} > .001$. The seven qualifying positions are five unique requests: OBS at $\Lambda = 1$ repeats across tolerances; group Fusion passes at $\Lambda = 1$ across tolerances and at $\Lambda = 1.5, \rho = 0$. The original fifteen candidates certified none. Each declaration has its own coverage guarantee.

A redesign under stronger assumptions ($\Lambda = 1.5, \rho = 0$). The jacket-type certificate moves all 28 vests' uniform mass to item 0, a men's bomber: its probability rises from $1/300$ to $29/300$; vests receive zero and 271 other items retain $1/300$. This exact reallocation lies outside all 15 original and 50 extended catalogue policies. The contrast is $(28/300)(m_0 - \overline{m}_{\text{vest}})$, where $\overline{m}_{\text{vest}}$ is the 28 vests' uniform mean. Its checked lower bound is $(28/300)(\ell_0 - b_{\text{vest}}/\kappa_{\text{vest}})$: row mass κ converts the group constraint to its uniform mean. The only nonzero terms, 2.472638 pp and -2.226283 pp, sum to 0.246355 pp in expected high-rating probability over uniform. This guarantees at least 2.46 additional expected high ratings per 1,000 assignments under the stated declaration. It exceeds 0.1 pp and matches the overlapping view's capacity; matched full-budget OBS and both R views have zero capacity.

Separate gender, jacket-type and color partitions give 144 view memberships: 10 exceed the threshold, 10 have positive capacity at most the threshold, and 124 have zero capacity. Construction and exact LP agree on 95 unique supported requests across the full evaluation; LP also resolves 30 overlaps. Subview capacity cannot exceed its parent's, as relaxation requires.

Under full-cell transport, each user's score is $Z_u = (300/16) \sum_{a \in S_u} c_a Y_{ua}$, with random panel S_u, binary high-rating outcomes Y_{ua} and policy contrast $c = \pi - \pi_0$. The factor $300/16$ corrects each item's inclusion probability $16/300$. A one-sided Hoeffding bound across 290 users uses score

range $[\min(0, 300 \min c), \max(0, 300 \max c)]$. The primary estimated gain over uniform is 0.371 pp, with lower bound -6.452 pp; none of 15 fixed queries passes. These bounds cover each prespecified query separately.

a) Catalogue queries with ordinary caching: A catalogue query tests one listed policy at a requested margin. At primary $(\Lambda, \rho) = (2, 0)$, each of five source views receives an extended catalogue combining two indicators: $\pi_i \propto t_1^{h_{1i}} t_2^{h_{2i}}$, $(t_1, t_2) \in \{.5, 1, 2\}^2 \setminus \{(1, 1)\}$. Ten pairs give 80 policy positions at margins $0, .001, .01$ (0, 0.1, 1 pp): 240 requests per view, 50 objectives and 150 objective–margin pairs. Prepared execution retains exact dual terms; optimized-exact adds sparse arithmetic and cached number conversions. Both fully cache objectives and retain a persistent checker. Native hot-start retains a solver basis without cross-objective certificate candidates. Quality checks precede three timing rounds.

Prepared and optimized-exact agree on all 1,200 requests. Optimized/prepared cost ratios 0.960, 0.941, 0.953 miss the ten-percent target; optimized-exact is slower in 3/15 view-rounds and retains more estimated exact-state payload in every view (not isolated RSS). Native hot-start changes 98 false answers to unknown and four unknown answers to false, never reversing a Boolean answer. These plans settle different query sets; on R projected, they agree and native is faster in all rounds. None certifies improvement, and neither registered fusion-only information target is supported. The constant-action witnesses already establish zero capacity.

b) The cost of family preprocessing: We test preprocessing on two 16-row jacket-type Fusion views, $(\Lambda, \rho) = (1.5, 0)$ and $(2, 0)$, selected retrospectively with known capacities. Table V compares Hot, Opt, and Opt with partition (Part) or exact-LP (LP) preprocessing. All share objective caching, checked points and an exact checker. Family certificates add only feasible points; a shared constant-action check skips unnecessary family solves. Stored-contrast and ideal-policy queries share exact fallback evidence.

All 32 stream executions (two views, four plans, one quality pass plus three timing rounds) complete. Each returns 240 false answers under both stored and ideal policies: no listed candidate meets the guarantee. At nonzero capacity, Part adds 5.1–7.7% to Opt's full cost without reducing its two native and two exact calls (Table V); LP saves no calls either. At zero capacity, family computation is skipped. The three rounds have incompletely balanced execution positions and different optimal points; costs compare complete preprocessing choices. Family answers enable redesign, without amortizing preprocessing here.

TABLE V
PREPROCESSING THE UNCHANGED 240-QUERY COAT CATALOGUE.

Plan	Full-stream time (s)	Cost relative to Opt	Solver calls N/E/F
Positive capacity $\Lambda = 1.5$			
Hot	6.800	0.974–1.009	2/2/0
Opt	6.934	1.000	2/2/0
Opt+Part	7.312	1.051–1.077	2/2/1
Opt+LP	7.276	1.042–1.079	2/2/1
Zero capacity $\Lambda = 2$			
Hot	6.329	0.985–0.998	1/1/0
Opt	6.390	1.000	1/1/0
Opt+Part	6.431	0.999–1.024	1/1/0
Opt+LP	6.359	0.990–1.002	1/1/0

Each stream has 240 queries. Time: three-round median including startup, preprocessing, verification and cleanup; relative cost: paired range. N/E/F: native/exact-policy/family solver calls per stream, identical across runs. Hot: native hot-start; Opt: optimized threshold execution. Both views use $\rho = 0$.

VII. Related Work and Scope

Database queries. Causal systems adjust aggregates [4], implement matching [30], define units [31] and explain patterns [32]. Uncertainty-annotated databases propagate bounds [5]; semantic views certify predicates using fresh draws [33]. Q-BRIDGE materializes source-dependent constraints once and evaluates changing policy objectives over their shared feasible set. Its additional query is whether any policy can meet a guarantee; matching certificates verify that answer and reusable terms reduce repeated arithmetic.

Our view combines binary density-ratio and conditional-moment relaxations [7], [8]. Other fusion methods exploit overlap [34], impose experimental moments [35], or estimate sensitivity frontiers [3]. Propensity-odds policy learning [2], [36] makes different restrictions from our outcome-conditional weights.

Classical allocation. Continuous maximin sharing [37, Sec. 2.2] allocates a common resource across groups. Our groups instead impose disjoint, two-sided constraints on uncertain means; we minimize the best action's advantage over a reference and recover an original-constraint certificate.

Proofs and scope. Filtering [38], assignment caching [39], bounds and reuse are established; VIPR checks exact optimization proofs [15]. IntegriDB authenticates results [40]; our digests bind certificates to a view. Our guarantees cover binary outcomes and known target weights under the stated source assumptions and intervals.

Data ethics. We reanalyze public benchmarks under their release terms, without recruiting or contacting participants. No institutional ethics review was undertaken or additional consent sought.

VIII. Conclusion

Q-BRIDGE turns integrated, uncertain evidence into a reusable object for policy queries. One coverage event supports changing objectives; matching certificates establish whole-family answers, and exact-term reuse reduces repeated-query cost. Coat distinguishes stopping on an exhausted view from redesign under an explicitly stronger declaration.

Acknowledgments and AI Usage Declaration

The authors conceived the research, designed the overall study, and led the manuscript writing. OpenAI Codex assisted with drafting, polishing and mathematical exposition (Sections I–VIII), code and experiments (Sections III–VI), figures, and source retrieval. The authors take full responsibility for the manuscript, including any text generated or polished by the LLM, and for ensuring its accuracy, originality, proper attribution, and compliance with publication ethics, including the prevention of plagiarism and scientific misconduct.

References

[1] T. Schnabel, A. Swaminathan, A. Singh, N. Chandak, and T. Joachims, "Recommendations as treatments: Debiasing learning and evaluation," in *Proceedings of the 33rd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 48. PMLR, 2016, pp. 1670–1679. [Online]. Available: https://proceedings.mlr.press/v48/schnabel16.html

[2] N. Kallus and A. Zhou, "Confounding-robust policy improvement," in *Advances in Neural Information Processing Systems*, vol. 31, 2018. [Online]. Available: https://proceedings.neurips.cc/paper/2018/hash/3a09a524440d44d7f19870070a5ad42f-Abstract.html

[3] Q. Lanners, C. Rudin, A. Volfovsky, and H. Parikh, "Data fusion for partial identification of causal effects," in *Advances in Neural Information Processing Systems*, vol. 38. Curran Associates, Inc., 2025, pp. 150 638–150 702. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2025/hash/dd4e79c2228c43954cc074abf2f85a07-Abstract-Conference.html

[4] B. Salimi, J. Gehrke, and D. Suciu, "Bias in OLAP queries: Detection, explanation, and removal," in *Proceedings of the 2018 International Conference on Management of Data*. ACM, 2018, pp. 1021–1035. [Online]. Available: https://doi.org/10.1145/3183713.3196914

[5] S. Feng, A. Huber, B. Glavic, and O. Kennedy, "Uncertainty annotated databases—a lightweight approach for approximating certain answers," in *Proceedings of the 2019 International Conference on Management of Data*. Association for Computing Machinery, 2019, pp. 1313–1330. [Online]. Available: https://www.cs.uic.edu/~bglavic/dbgroup/assets/pdfpubls/FH19.pdf

[6] J. Fawkes, M. O'Riordan, A. Vlontzos, O. Corcoll, and C. M. Gilligan-Lee, "The hardness of validating observational studies with experimental data," in *Proceedings of the 28th International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 258. PMLR, 2025, pp. 1819–1827. [Online]. Available: https://proceedings.mlr.press/v258/fawkes25b.html

[7] J. Dorn, K. Guo, and N. Kallus, "Doubly-valid/doubly-sharp sensitivity analysis for causal inference with unmeasured confounding," *Journal of the American Statistical Association*, vol. 120, no. 549, pp. 331–342, 2025. [Online]. Available: https://doi.org/10.1080/01621459.2024.2335588

[8] K. Ishikawa, N. He, and T. Kanamori, "A convex framework for confounding robust inference," *Journal of Machine Learning Research*, vol. 27, no. 62, pp. 1–53, 2026. [Online]. Available: https://jmlr.org/papers/v27/23-1434.html

[9] C. J. Clopper and E. S. Pearson, "The use of confidence or fiducial limits illustrated in the case of the binomial," *Biometrika*, vol. 26, no. 4, pp. 404–413, 1934. [Online]. Available: https://doi.org/10.1093/biomet/26.4.404

[10] A. Maurer and M. Pontil, "Empirical Bernstein bounds and sample variance penalization," in *Proceedings of the 22nd Annual Conference on Learning Theory*, Montreal, Canada, 2009. [Online]. Available: https://www.cs.mcgill.ca/~colt2009/papers/012.pdf

[11] R. Berk, L. Brown, A. Buja, K. Zhang, and L. Zhao, "Valid post-selection inference," *The Annals of Statistics*, vol. 41, no. 2, pp. 802–837, 2013. [Online]. Available: https://arxiv.org/abs/1306.1059

[12] D. Bertsimas, V. Gupta, and N. Kallus, "Data-driven robust optimization," *Mathematical Programming*, vol. 167, no. 2, pp. 235–292, 2018.

[13] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge University Press, 2004. [Online]. Available: https://web.stanford.edu/~boyd/cvxbook/

[14] A. Neumaier and O. Shcherbina, "Safe bounds in linear and mixed-integer linear programming," *Mathematical Programming*, vol. 99, no. 2, pp. 283–296, 2004. [Online]. Available: https://arnold-neumaier.at/ms/mip.pdf

[15] K. K. H. Cheung, A. Gleixner, and D. E. Steffy, "Verifying integer programming results," in *Integer Programming and Combinatorial Optimization*, ser. Lecture Notes in Computer Science, vol. 10328. Springer, 2017, pp. 148–160. [Online]. Available: https://arxiv.org/abs/1611.08832

[16] M. Sion, "On general minimax theorems," *Pacific Journal of Mathematics*, vol. 8, no. 1, pp. 171–176, 1958. [Online]. Available: https://msp.org/pjm/1958/8-1/pjm-v8-n1-p14-s.pdf

[17] H. A. Eiselt, "Continuous maximin knapsack problems with GLB constraints," *Mathematical Programming*, vol. 36, pp. 114–121, 1986.

[18] T. Kuno, H. Konno, and E. Zemel, "A linear-time algorithm for solving continuous maximin knapsack problems," *Operations Research Letters*, vol. 10, no. 1, pp. 23–26, 1991.

[19] T. Marcucci and R. Tedrake, "Warm start of mixed-integer programs for model predictive control of hybrid systems," *IEEE Transactions on Automatic Control*, vol. 66, no. 6, pp. 2433–2448, 2021. [Online]. Available: https://groups.csail.mit.edu/robotics-center/public_papers/Marcucci19.pdf

[20] H. Kothari and J. R. Luedtke, "Accelerating Benders decomposition for solving a sequence of sample average approximation replications," *Mathematical Programming Computation*, 2026, published online May 23, 2026. [Online]. Available: https://link.springer.com/article/10.1007/s12532-026-00324-3

[21] A. Deza, F. Badilla, and S. S. Dey, "Mixed-integer program sensitivity analysis via branch-and-bound trees," 2026, preprint, version 1, submitted November 3, 2026. [Online]. Available: https://arxiv.org/abs/2611.04415v1

[22] HiGHS developers, "HiGHS documentation: Hot start and python interfaces," Official software documentation, 2026, accessed 2026-11-04; API cross-checked with HiGHS v1.12.0 source and local HiGHS 1.12.0 and 1.15.1 bindings. [Online]. Available: https://ergo-code.github.io/HiGHS/dev/guide/further/

[23] ——, "HiGHS documentation: List of options," 2026, official documentation, accessed 2026-11-04; option semantics checked against local HiGHS 1.15.1. [Online]. Available: https://ergo-code.github.io/HiGHS/dev/options/definitions/

[24] S. Martello and P. Toth, *Knapsack Problems: Algorithms and Computer Implementations*, ser. Wiley-Interscience Series in Discrete Mathematics and Optimization. Chichester, UK: John Wiley & Sons, 1990. [Online]. Available: https://silvano333.github.io/kp.html

[25] Q. Huangfu and J. A. J. Hall, "Parallelizing the dual revised simplex method," *Mathematical Programming Computation*, vol. 10, no. 1, pp. 119–142, 2018. [Online]. Available: https://doi.org/10.1007/s12532-017-0130-5

[26] Zuse Institute Berlin, "SoPlex 8.1.0," https://soplex.zib.de/, 2026, released September 2026. Accessed November 6, 2026.

[27] A. M. Gleixner, D. E. Steffy, and K. Wolter, "Iterative refinement for linear programming," *INFORMS Journal on Computing*, vol. 28, no. 3, pp. 449–464, 2016.

[28] A. Gleixner and D. E. Steffy, "Linear programming using limited-precision oracles," *Mathematical Programming*, vol. 183, pp. 525–554, 2020.

[29] C. Gao, S. Li, W. Lei, J. Chen, B. Li, P. Jiang, X. He, J. Mao, and T.-S. Chua, "KuaiRec: A fully-observed dataset and insights for evaluating recommender systems," in *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*. ACM, 2022, pp. 540–550. [Online]. Available: https://kuairec.com/

[30] B. Salimi, C. Cole, D. R. K. Ports, and D. Suciu, "ZaliQL: Causal inference from observational data at scale," *Proceedings of the VLDB Endowment*, vol. 10, no. 12, pp. 1957–1960, 2017. [Online]. Available: https://www.vldb.org/pvldb/vol10/p1957-salimi.pdf

[31] B. Salimi, H. Parikh, M. Kayali, L. Getoor, S. Roy, and D. Suciu, "Causal relational learning," in *Proceedings of the 2020 ACM SIGMOD International Conference on Management of Data*. ACM, 2020, pp. 241–256. [Online]. Available: https://doi.org/10.1145/3318464.3389759

[32] B. Youngmann, M. Cafarella, A. Gilad, and S. Roy, "Summarized causal explanations for aggregate views," *Proceedings of the ACM on Management of Data*, vol. 2, no. 1, pp. 71:1–71:27, 2024. [Online]. Available: https://doi.org/10.1145/3639328

[33] O. Hassanzadeh and D. Subramanian, "Semantic views: Certified reuse of LLM labels across semantic queries," 2026, preprint, version 1, submitted October 8, 2026. [Online]. Available: https://arxiv.org/abs/2610.12768v1

[34] K. Jiang, W. Hu, X. Lai, S. Yang, and X.-H. Zhou, "Improving sensitivity analysis by synthesizing randomized clinical trials with limited overlap," *Statistics in Medicine*, vol. 45, no. 18–19, p. e70637, 2026. [Online]. Available: https://onlinelibrary.wiley.com/doi/10.1002/sim.70637

[35] Y. Du, Z. Zhang, H. Li, C. Fang, J. Xu, Z. Peng, and J. Guo, "Feasible fusion: Constrained joint estimation under structural non-overlap," in *Proceedings of the 43rd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 306. PMLR, 2026, pp. 26 851–26 885. [Online]. Available: https://proceedings.mlr.press/v306/du26s.html

[36] K. Hess, D. Frauen, V. Melnychuk, and S. Feuerriegel, "Efficient and sharp off-policy learning under unobserved confounding," in *International Conference on Learning Representations*, 2026, pp. 57 552–57 587. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2026/file/5e972fe492946fc7a02aef64d0f4fd7e-Paper-Conference.pdf

[37] T. Yamada, M. Futakawa, and S. Kataoka, "Some exact algorithms for the knapsack sharing problem," *European Journal of Operational Research*, vol. 106, no. 1, pp. 177–183, 1998. [Online]. Available: https://www.nda.ac.jp/~yamada/paper/98-2.pdf

[38] T. Bartels, V. Fisikopoulos, and M. Weiser, "Fast floating-point filters for robust predicates," *BIT Numerical Mathematics*, vol. 63, 2023, article 31. [Online]. Available: https://doi.org/10.1007/s10543-023-00975-x

[39] C. Cadar, D. Dunbar, and D. Engler, "KLEE: Unassisted and automatic generation of high-coverage tests for complex systems programs," in *USENIX Symposium on Operating Systems Design and Implementation*, 2008. [Online]. Available: https://static.usenix.org/event/osdi08/tech/full_papers/cadar/cadar_html/paper.html

[40] Y. Zhang, J. Katz, and C. Papamanthou, "IntegriDB: Verifiable SQL for outsourced databases," in *Proceedings of the 22nd ACM SIGSAC Conference on Computer and Communications Security*. Association for Computing Machinery, 2015, pp. 1480–1491. [Online]. Available: https://integridb.github.io/IntegriDB.pdf